

## 1626 Brackets sequence

Let us define a regular brackets sequence in the following way:

1. Empty sequence is a regular sequence.
2. If S is a regular sequence, then (S) and [S] are both regular sequences.
3. If A and B are regular sequences, then AB is a regular sequence.

For example, all of the following sequences of characters are regular brackets sequences:

() , [] , (()) , ([ ]), () [] , () [()]

And all of the following character sequences are not:

( , [ , ) , )( , ([ ]), ([

Some sequence of characters '(', ')', '[', and ']' is given. You are to find the shortest possible regular brackets sequence, that contains the given character sequence as a subsequence. Here, a string  $a_1a_2 \dots a_n$  is called a subsequence of the string  $b_1b_2 \dots b_m$ , if there exist such indices  $1 \leq i_1 < i_2 < \dots < i_n \leq m$ , that  $a_j = b_{i_j}$  for all  $1 \leq j \leq n$ .

### Input

The input begins with a single positive integer on a line by itself indicating the number of the cases following, each of them as described below. This line is followed by a blank line, and there is also a blank line between two consecutive inputs.

The input file contains at most 100 brackets (characters '(', ')', '[', and ']') that are situated on a single line without any other characters among them.

### Output

For each test case, the output must follow the description below. The outputs of two consecutive cases will be separated by a blank line.

Write to the output file a single line that contains some regular brackets sequence that has the minimal possible length and contains the given sequence as a subsequence.

### Sample Input

1

([([

### Sample Output

() [()]